Advances in Multimodal Retrieval-Augmented Generation Systems

Abstract
This paper presents a high-level overview of Multimodal Retrieval-Augmented Generation (RAG) systems that integrate textual and visual information for intelligent document understanding. The study discusses modern architectures, embedding techniques, vector databases, and large language models used in enterprise knowledge retrieval systems.

1. Introduction

Multimodal AI systems are transforming document intelligence by enabling unified retrieval across text, images, charts, and tables.

2. Background

Recent advances in Large Language Models (LLMs), CLIP embeddings, and vector databases have accelerated the adoption of multimodal search systems.

3. Methodology

A typical pipeline includes document ingestion, text and image extraction, embedding generation, vector indexing, retrieval, and response generation.

4. Applications

Enterprise search, healthcare, legal document analysis, technical documentation retrieval, and research assistance.

5. Future Work

Hybrid search, agentic workflows, multimodal reasoning, and scalable cloud-native deployment architectures.

6. Conclusion

Multimodal RAG systems represent a significant step toward intelligent knowledge discovery and context-aware AI assistants.

References

[1] Research literature on LLMs, CLIP embeddings, vector search, and multimodal retrieval systems.